

Graphical Processing Systems

Lectures and practical works

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Course Syllabus

- ❑ Hidden line and surface presentation
- ❑ 3D object representation
- ❑ Polygonal objects tracing. Coordinate systems
- ❑ Local reflection model: Phong model
- ❑ Illumination models
- ❑ Shadow generation techniques
- ❑ Textures. Mapping techniques
- ❑ Global reflection models: Ray-tracing, Radiosity
- ❑ Animation techniques
- ❑ Graphics formats
- ❑ Graphical processing units
- ❑ Parallel graphics visualization
- ❑ Graphics application development environments
- ❑ Graphics libraries (OpenGL, DirectX, ...)
- ❑ Graphics languages (SVG, VRML, XML, HTML, WebGL)

Laboratory Syllabus

- ❑ Mathematics for computer graphics
- ❑ Basic structure of the Open GL application
- ❑ Graphics pipeline
- ❑ Basic vertex and fragment shaders
- ❑ Graphics primitives and transformations in OpenGL
- ❑ Graphics models, file formats
- ❑ Projections, clipping planes
- ❑ illumination models, texture mapping, and shadow computation
- ❑ Environment mapping
- ❑ Ray tracing algorithm
- ❑ Bump mapping

References

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- 2. Pharr M., Humphreys G.: *"Physically-Based Rendering"*, Morgan Kaufmann, 2nd edition, 2010.
- 3. Shreiner D., *"OpenGL programming guide : the official guide to learning OpenGL, versions 3.0 and 3.1"*, 7th Edition, 2010.
- 4. Vince J.: *"3-D Computer Animation"*, Addison-Wesley Publ., 1992.
- 5. Watt A., Policarpo F.: *"3D Games. Real-time Rendering and Software Technology"*. Addison-Wesley, 2001.
- 6. Watt A.: *"3D Computer Graphics"*. Addison-Wesley, 3rd Edition, 2013.

References

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2. CGAL - [Computational Geometry Algorithms Library](#)
3. [Indispensable Computer Graphics Sites](#)
4. dmoz - [Algorithms and Data Structures](#)
5. [Graphics Libraries](#)
6. [Dictionary of Algorithms and Data Structures](#)
7. [NVIDIA Technology](#)